

SLIP – 16

Section I: Data Structures [15 Marks]

Q1) Read the data from file 'cities.txt' containing names of 10 cities and their STD codes. Accept a name of the city from user and use linear search algorithm to check whether the name is present in the file and output the STD code, otherwise output “city not in the list”. i) Accept & Display Function ii) Linear Search Function

Assumed file format (one record per line): CityName STDcode e.g. Pune 020

Answer:

```
#include <stdio.h>
#include <string.h>

struct City
{
    char name[50];
    char std[10];
};

int accept(struct City c[])
{
    FILE *fp;
    int n = 0;

    fp = fopen("cities.txt", "r");
    if(fp == NULL)
    {
        printf("File not found\n");
        return 0;
    }

    while(n < 10 && fscanf(fp, "%s %s", c[n].name, c[n].std) == 2)
        n++;
    fclose(fp);
    return n;
}

void display(struct City c[], int n)
{
    int i;

    for(i = 0; i < n; i++)
        printf("%s\n", c[i].name, c[i].std);
}
```

```

int linearSearch(struct City c[], int n, char key[])
{
    int i;

    for(i = 0; i < n; i++)
    {
        if(strcmp(c[i].name, key) == 0)
            return i;
    }
    return -1;
}

int main()
{
    struct City c[10];
    char key[50];    int n,
    pos;

    n = accept(c);
    if(n == 0)
        return 0;

    display(c, n);

    printf("Enter city name: ");
    scanf("%s", key);

    pos = linearSearch(c, n, key);

    if(pos != -1)
        printf("STD code of %s is %s",
key, c[pos].std);
    else
        printf("city not in the
list");

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (primary keys are underlined).

Student (sno ,s_name, s_class)

Teacher (tno ,t_name, yrs_experience)

The relationship is as follows:

Student-Teacher: M-M with descriptive attribute Subject.

Constraint: s_class should be either "FY", "SY" or "TY"

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Student(  
sno INT PRIMARY KEY,  
s_name VARCHAR(50),  
s_class VARCHAR(2)  
CHECK(s_class IN ('FY','SY','TY'))  
);  
  
CREATE TABLE Teacher(  
tno INT PRIMARY KEY,  
t_name VARCHAR(50),  
yrs_experience INT  
);  
  
CREATE TABLE Student_Teacher(  
sno INT REFERENCES Student(sno) ON DELETE CASCADE,  
tno INT REFERENCES Teacher(tno) ON DELETE CASCADE,  
subject VARCHAR(50),  
PRIMARY KEY(sno, tno, subject)  
);  
  
INSERT INTO Student VALUES  
(101,'Amit','FY'),  
(102,'Neha','SY'), (103,'Raj','TY'),  
(105,'Sita','TY');  
  
INSERT INTO Teacher VALUES  
(1,'Pawar',8),  
(2,'Joshi',3),  
(3,'Kale',12);  
  
INSERT INTO Student_Teacher VALUES  
(101,1,'Maths'),  
(102,2,'Data Structure'),  
(103,1,'Data Structure'),  
(105,3,'Maths'),  
(103,3,'DBMS');
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. List teachers who has more than 5 years of experience.

Answer:

```
SELECT * FROM Teacher
WHERE yrs_experience > 5;
```

ii. Count all students who have taken subject "Data Structure"

Answer:

```
SELECT COUNT(DISTINCT sno)
FROM Student_Teacher
WHERE subject = 'Data Structure';
```

iii. List the names of all students from SY Class.

Answer:

```
SELECT s_name FROM Student WHERE
s_class = 'SY';
```

iv. Change experience of teacher named "Pawar" to 20.

Q 2(C)

C. Write a query to list the names of all teachers with their subjects along with the student names to whom they are teaching. [4 marks]

Answer:

```
SELECT T.t_name, ST.subject, S.s_name
FROM Teacher T
JOIN Student_Teacher ST ON T.tno = ST.tno
JOIN Student S ON S.sno = ST.sno;
```

Q 3) Viva [5 marks]